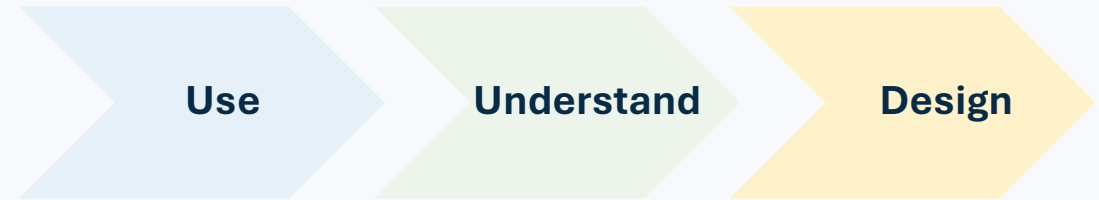


CompSci 471

Artificial Intelligence

Lesson 1: Intro to Artificial Intelligence

Using AI → Understanding AI → Designing AI



**Learn to question AI
before you rely on it.**



Introductions

A little about me....

Maj Tiffany M. Phan

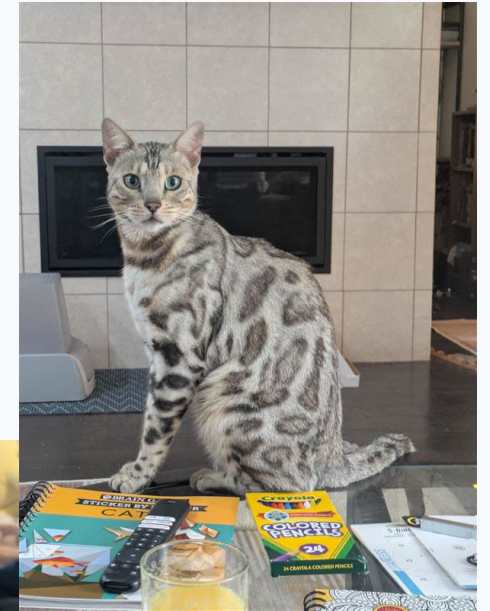
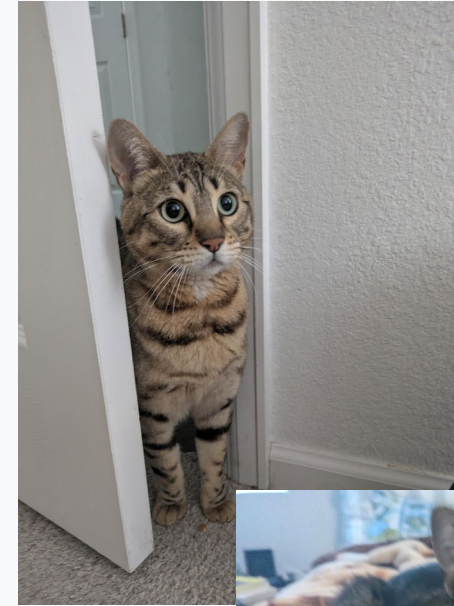
Office: [Madera 405 \(Mezzanine level\)](#)

Email: tiffany.phan@afacademy.af.edu

Preferred Communications: [Microsoft Teams](#)

Office Hours – check my Outlook calendar

El Preference – message me to schedule



Today's mission

By the end of Lesson 1, students should be able to...

1

Describe what AI systems do using concrete examples

2

Identify strengths and limitations of modern AI systems

3

Explain why AI outputs require evaluation and cannot be blindly trusted

Main habit: ask “how do I know?” before acting on an AI output.

Course startup

Get oriented with the course.

Before we start using AI as a case study:

Blackboard enrollment

Confirm you can see course announcements, assignments, and quizzes.

Syllabus

Know the grading structure, AI-use policy, and assessment rhythm.

GitHub Teachbook / slides

This is where the open course materials will live.

Default assumption: Level 0 GenAI unless your instructor says otherwise.

Live demo

Let's ask an AI system a question. Your job is to decide whether you trust it.

Prompt to try

Decide whether an autonomous drone should fly directly through a mountain pass or take a longer route around it. Make a single decision.

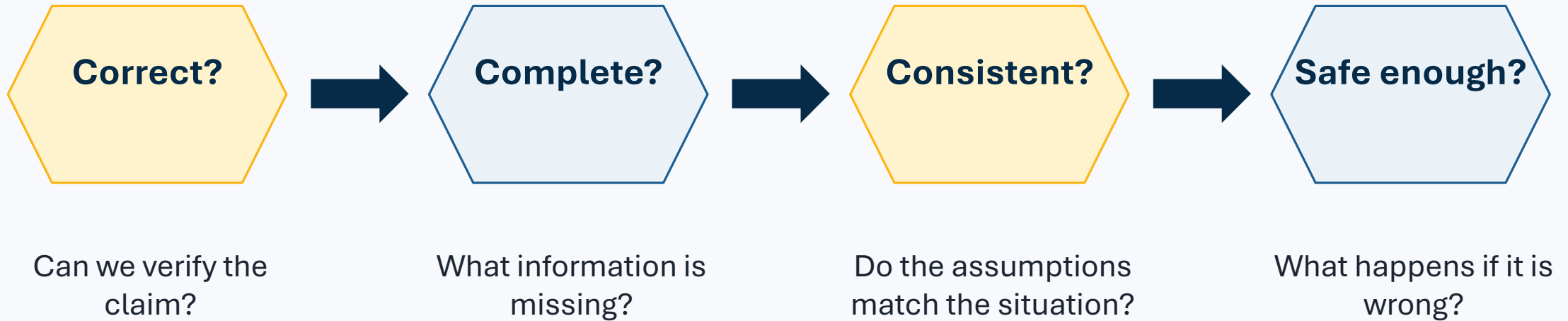
Watch for

Does it invent missing information?
Does it sound more certain than the evidence allows?
Does it mention verification or risk?
What would a real system need before acting?

You are not evaluating style. You are evaluating trust.

The trust test

A quick frame for evaluating any AI output.



Class question: which step is easiest to skip when an answer sounds fluent?

What just happened?

AI can look intelligent before it is reliable.

The confidence trap

A system can produce an answer that is fluent, specific, and useful-looking...



...while still being incomplete, misleading, or wrong.

**The lesson is not
“AI is bad.”**

**The lesson is
“AI requires evaluation.”**

What is artificial intelligence?

For this course, define AI through what systems are designed to do.

AI systems are designed to...

Solve problems

find routes, schedules, plans

Learn from data

detect patterns and update models

Make decisions

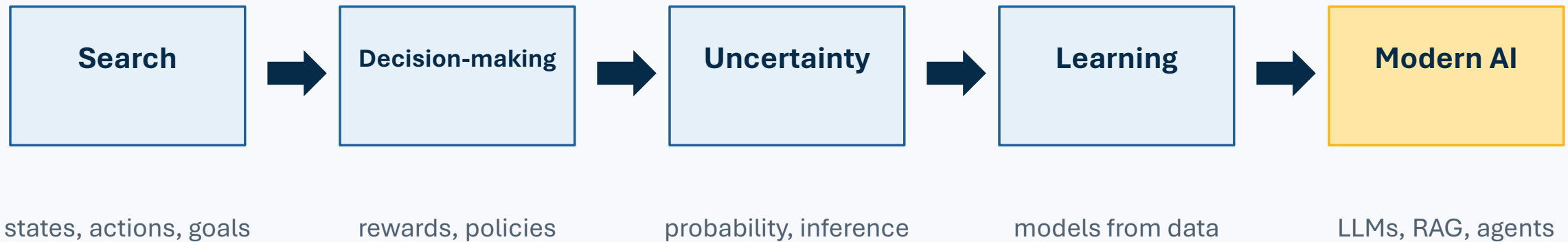
select actions under constraints

Reason under uncertainty

estimate what is likely or risky

AI is more than LLMs

Large language models are one important type of AI, not the whole field.



Course theme: select the right AI technique for the problem, then evaluate its behavior.

Examples of AI in the real world

Which of these feels the most intelligent? Which would you trust the least?

Drone route planning

Cyber alert triage

**AI
system**

**Aircraft maintenance
prediction**

Medical image support

Language assistant

Strengths of AI

Modern AI systems are powerful when the task structure fits their strengths.

Scale



process more data than humans can inspect manually

Patterns



find statistical structure in noisy inputs

Speed



generate candidate answers quickly

Repetition



apply the same rule or model consistently

Strength $\neq$ trust. It means the system may be useful under the right constraints.

Limitations of AI

The same systems can fail in ways that are hard to notice.

Plausibility vs. truth

An answer can sound right without being right.

Missing grounding

The model may not know what is physically or operationally true.

Input sensitivity

Small wording changes can shift the output.

Hallucination

It may invent facts, sources, or constraints.

Edge cases

It can behave poorly outside familiar patterns.

Unstated assumptions

It may hide uncertainty behind confident language.

Evaluation checklist

Use this when an AI answer “looks good.”

Before acting on an AI output, ask:

- | | |
|------------------------|---|
| 1. Logic | Is the reasoning internally consistent? |
| 2. Evidence | Can the claims be verified? |
| 3. Assumptions | What had to be assumed? |
| 4. Missing info | What did the answer ignore? |
| 5. Stakes | What is the cost of being wrong? |

This checklist will come back throughout the course.

Mini-debrief: spot the risk

Use the checklist on a concrete AI-style recommendation.

AI recommendation: “Take the direct drone route through the pass. It is the fastest and therefore the best option.”

What should make us pause?

Fastest is not always safest or best. Weather, battery, terrain, and rules are missing.

“Best” depends on the objective and constraints.

A real system would need verification data.

AI answers often skip the problem formulation step. That is where Lesson 4 begins.

AI and humans are different tools

The goal is not replacement; it is responsible system design.

Task	AI	Human
Pattern recognition	Often strong	Often variable
Context understanding	Often weak	Often strong
Consistency	Often strong	Can vary
Judgment under stakes	Needs design + oversight	Needs training + accountability

Discussion: where should we rely on AI, where should we verify, and where should we not automate?

Course overview

The semester turns AI from a black box into a set of design choices.

Represent

— states, actions, goals

Search & decide

— frontiers, heuristics, games

Reason under uncertainty

— probability, Bayes, temporal models

Learn from data

— classifiers, neural networks, deep learning

Build modern systems

— LLMs, RAG, agents, ethics

Using AI → Understanding AI → Designing AI

Takeaways

What you should leave with today.

AI is powerful, useful, and imperfect.

Concrete examples matter

AI is best understood by what the system is designed to do.

Strengths and limits travel together

The same capability can create risk when the context changes.

Evaluation is a skill

Do not confuse confident output with verified knowledge.

Next class: Anatomy of AI Systems — what is actually inside an AI system?